

Jesse Zeng

Toronto, ON | jesse.zeng.work@gmail.com | github.com/Jesse-Zeng423 | linkedin.com/in/jesse-zeng

Computer Science undergraduate developing reliable AI applications across web, browser agents, and smart devices.

EDUCATION

University of Toronto, St. George

Sep 2025 – Apr 2029 (expected)

Honours Bachelor of Science, Computer Science

Johns Hopkins University

Jul 2024 – Aug 2024

Summer Program, Baltimore, MD: Analyzed obesity risk factors with Python and XGBoost

EXPERIENCE

AI & Cloud Solutions Engineering Intern, Project Team Lead

Jul 2026 – Aug 2026

Aspire Digital Technology (Shenzhen) Co., Ltd., Guangzhou Branch

- Led development of a Rokid AR guide that helps visitors identify exhibits and explore them through questions on smart glasses.
- Integrated the Android camera pipeline with a Python service and prioritized fresh frames to make exhibit recognition more responsive.
- Grounded exhibit-specific answers in reviewed knowledge and separated verified introductions from AI-generated Q&A to make explanations more dependable.
- Refined on-glasses text and speech for visitors; recognition, exhibit switching, and introduction playback passed user acceptance and regression checks.
- Evaluated AI agents on search, tool use, and scheduled tasks; documented reproducible failures to inform reliability decisions. Also improved task feedback and exception alerts in frontend and mini-program features.

PROJECTS

AllAbout Campus

Sep 2026

Team Lead, Battle of Schools Hackathon

Tech: React, TypeScript, Fastify, Steel SDK, Playwright

- Directed a hackathon team to deliver a campus web-agent demo for U of T and Waterloo questions, returning answers with cited sources.
- Unified frontend, backend, and live browser workflows; implemented source routing, evidence extraction, and coverage reporting to surface inaccessible or insufficient sources.
- Added preflight and regression checks for routing, browser execution, cancellation, cleanup, and evidence integration to make the public demo repeatable.

Memory Applicability Guard

May 2026 – Aug 2026

Independent Project

Tech: Python, structured JSON, unit testing

- Created an open-source prototype that helps AI agents decide when to use, ignore, or clarify stored memory, guarding against revoked permissions, stale facts, and evidence gaps.
- Packaged the decision helper as an installable agent skill and verified its behavior with deterministic tests.

FlowList

Apr 2026

Independent Project

Tech: Next.js, React, TypeScript, YouTube Data API, Vercel

- Launched a playlist-sequencing app that imports public YouTube playlists or accepts manual entries, giving listeners a way to shape more coherent music flows.
- Prototyped mood chapters, energy curves, and transition explanations from track metadata while keeping API credentials server-side.

VOLUNTEERING

AI Education Volunteer

Sep 2024 – Dec 2024

Rural Education Program

- Introduced rural learners in Xinshao, China to computers, classification, and neural networks through accessible lessons and hands-on activities, adapting the material to limited classroom technology.
- When the classroom had no computer for the slides I brought, switched to a Peppa Pig robot story to explain the basic idea of machine learning.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Kotlin, Java, R

Frameworks & libraries: React, Next.js, FastAPI, Fastify, pandas, NumPy, XGBoost

Tools & platforms: Git, Playwright, Steel SDK, Android, Vercel, Dify, BLE, MediaCodec